NATIONAL AERONAUTICS AND SPACE ADMINISTRATION

MISSION CONTROL TECHNICAL REFERENCE

ODYSSEY SPACECRAFT — EMERGENCY SYSTEMS MANUAL

DOCUMENT NUMBER	DOC-MC-2087-TC-001	**REVISION**	REV C
DATE OF ISSUE	2087.03.14	**CLASSIFICATION**	RESTRICTED
MISSION	ODYSSEY — EUROPA RETURN	**STATUS**	ACTIVE
PREPARED BY	Dir. S. Chen / Dr. M. Webb	**APPROVED BY**	NASA/ESA Joint Ops

■ CRITICAL SAFETY NOTICE

This document is intended EXCLUSIVELY for Mission Control Operators. The Astronaut must NOT have access to this manual at any time during the mission. Failure to maintain information asymmetry will result in mission abort. Handle with appropriate security protocols.

MANUAL SCOPE & CONTENTS

This technical reference manual provides Mission Control Operators with all procedures, reference tables, diagnostic trees, and resolution sequences required to guide the Astronaut through emergency system restoration aboard the Odyssey spacecraft. Modules covered in this edition: • MODULE A — Life Support Systems (Section 2) • MODULE B — Electrical Distribution System (Section 3) Each module section contains: system overview, fault identification tables, resolution procedures, and operator communication scripts.

QUICK REFERENCE

ENERGY LOST PER ERROR	10%	MAX ERRORS (NORMAL)	3	ENERGY BONUS / MODULE	+5%

SEC. 2 MODULE A — LIFE SUPPORT SYSTEMS

2.1 SYSTEM OVERVIEW

The Odyssey Life Support System (LSS) maintains breathable atmosphere within the crew compartment via a network of three independently controlled gas valves: OXYGEN INTAKE (O2), CARBON DIOXIDE SCRUBBER (CO2), and NITROGEN BALANCE (N2). Each valve is represented on the Astronaut's panel by a rotary dial with positions 0–9.

■ ASTRONAUT PANEL DESCRIPTION	The Astronaut sees three dials labeled O2, CO2 and N2, each showing a current reading from 0 to 9. Below the dials is a STATUS INDICATOR showing either STABLE (green), WARNING (yellow) or CRITICAL (red). There may also be a HULL TEMPERATURE display showing a value between 18°C and 45°C.

2.2 FAULT IDENTIFICATION

When the LSS module activates, the Astronaut must read out all three dial values and the STATUS INDICATOR colour. Use the tables below to determine the correct target values.

TABLE 2-A — Status Indicator Reference

STATUS	INDICATOR COLOR	ACTION REQUIRED	TIME LIMIT
STABLE	GREEN	Minor adjustment — follow TABLE 2-B	120 sec
WARNING	YELLOW	Moderate fault — follow TABLE 2-C	75 sec
CRITICAL	RED	Severe fault — follow TABLE 2-D IMMEDIATELY	45 sec

TABLE 2-B — STABLE Status: Target Values

Find the current O2 value in the left column. Read the target values for CO2 and N2 across.

O2 READING	TARGET O2	TARGET CO2	TARGET N2	NOTES
0–1	3	2	5	Low O2 — prioritise O2 dial first
2–3	3	3	4	Standard low — adjust CO2 first
4–5	4	3	3	Nominal range — minor balance
6–7	4	4	2	Elevated O2 — reduce O2 first
8–9	5	4	1	High O2 — follow in order: O2 → N2 → CO2

TABLE 2-C — WARNING Status: Target Values

Check HULL TEMPERATURE display first. Use the correct sub-table based on temperature range.

HULL TEMP 18–30°C

O2	TARGET O2	TARGET CO2	TARGET N2
0–2	4	2	4
3–5	4	3	3
6–7	5	3	2
8–9	5	4	1

HULL TEMP 31–45°C

O2	TARGET O2	TARGET CO2	TARGET N2
0–2	5	1	4
3–5	5	2	3
6–7	6	2	2
8–9	6	3	1

TABLE 2-D — CRITICAL Status: Emergency Resolution

■ CRITICAL FAULT — TIME LIMIT: 45 SECONDS Do NOT consult TABLE 2-B or 2-C during CRITICAL status. Use this table only. Instruct Astronaut to set ALL dials to target simultaneously, starting with CO2.

SCENARIO	CONDITION	TARGET O2	TARGET CO2	TARGET N2
Alpha	O2 < 3 AND CO2 > 6	5	2	3
Beta	O2 > 7 AND N2 < 2	4	3	3
Gamma	ALL values 0 OR ALL values 9	5	3	2
Delta	Any other combination	4	4	2

2.3 RESOLUTION PROCEDURE

1	**Ask O2 value**	Instruct Astronaut: 'Read O2 dial value.' Record the number.
2	**Ask CO2 value**	Instruct Astronaut: 'Read CO2 dial value.' Record the number.
3	**Ask N2 value**	Instruct Astronaut: 'Read N2 dial value.' Record the number.
4	**Ask Status Indicator**	Instruct Astronaut: 'What color is the Status Indicator?' — GREEN, YELLOW, or RED.
5	**Check Hull Temp (if WARNING)**	If STATUS is WARNING — Instruct Astronaut: 'What is the Hull Temperature reading?'
6	**Determine targets**	Using the appropriate table above, find the correct target values for O2, CO2, and N2.
7	**Give instructions**	State clearly: 'Set O2 to [X], CO2 to [Y], N2 to [Z].' Give one at a time if needed.
8	**Confirm resolution**	Module is solved when the STATUS INDICATOR turns GREEN and a confirmation tone sounds.

■ OPERATOR NOTE If the Astronaut reports difficulty reaching a dial value (e.g., dial stuck at 9 and target is 2), instruct them to rotate the dial COUNTER-CLOCKWISE past 0 and continue to the target. This wraps around correctly on all dial types.

SEC. 3 MODULE B — ELECTRICAL DISTRIBUTION SYSTEM

3.1 SYSTEM OVERVIEW

The Odyssey Electrical Distribution System (EDS) manages power flow through six circuit breakers labeled A through F. Each breaker has three states: ON (green LED), OFF (no LED), and TRIPPED (red LED flashing). The Astronaut must restore the correct configuration by toggling breakers in a specific sequence determined by the fault code displayed on the EDS control panel.

■ ASTRONAUT PANEL DESCRIPTION	The Astronaut sees 6 toggle switches labeled A, B, C, D, E, F with LEDs above each. Below the switches is a FAULT CODE display showing a 3-character alphanumeric code (e.g. '3B7' or 'A2F'). There is also a LOAD METER showing a percentage (0–100%).

3.2 FAULT CODE REFERENCE

The FAULT CODE determines which breakers need to be toggled and in what sequence. The code has three characters: [NUMBER][LETTER][NUMBER]. Each character modifies the procedure.

TABLE 3-A — First Character (Number) Modifier

1ST CHAR	BREAKERS TO TOGGLE OFF	BREAKERS TO KEEP ON
1	Toggle OFF: B, D	Keep ON: A, C, E, F
2	Toggle OFF: A, E	Keep ON: B, C, D, F
3	Toggle OFF: C, F	Keep ON: A, B, D, E
4	Toggle OFF: B, C, E	Keep ON: A, D, F
5	Toggle OFF: A, D, F	Keep ON: B, C, E
6	Toggle OFF: D, E	Keep ON: A, B, C, F
7	Toggle OFF: A, B, F	Keep ON: C, D, E
8	Toggle OFF: C, D	Keep ON: A, B, E, F
9	Toggle OFF: E, F	Keep ON: A, B, C, D

TABLE 3-B — Second Character (Letter) Sequence Modifier

The letter determines the ORDER in which to toggle the breakers identified in TABLE 3-A.

2ND CHAR	TOGGLE SEQUENCE	NOTES
A	**Alphabetical order (A → F)**	Toggle from lowest to highest labeled breaker
B	**Reverse alphabetical (F → A)**	Toggle from highest to lowest labeled breaker
C	**Odd-labeled first (A, C, E then B, D, F)**	Toggle odd positions, then even
D	**Even-labeled first (B, D, F then A, C, E)**	Toggle even positions, then odd
E	**As listed in TABLE 3-A, left to right**	Follow the exact order stated in the fault row
F	**Reverse of TABLE 3-A order, right to left**	Reverse the order stated in the fault row

TABLE 3-C — Third Character (Number) Load Check

After toggling breakers, check the LOAD METER. If the load does not match the acceptable range for the third character, one additional breaker toggle is required.

3RD CHAR	ACCEPTABLE LOAD RANGE	IF LOAD OUT OF RANGE — TOGGLE
1	10% – 25%	Toggle breaker A
2	26% – 40%	Toggle breaker B
3	41% – 55%	Toggle breaker C
4	56% – 65%	Toggle breaker D
5	66% – 75%	Toggle breaker E
6	76% – 82%	Toggle breaker F
7	83% – 88%	Toggle breaker A, then C
8	89% – 93%	Toggle breaker B, then D
9	94% – 100%	Toggle ALL remaining ON breakers OFF, then ON again

3.3 WORKED EXAMPLE

■ EXAMPLE FAULT CODE: '4CF'

1st char = 4 → Toggle OFF breakers B, C, E (TABLE 3-A) 2nd char = C → Toggle odd-labeled first: C (odd), then B, E (even) → So sequence is: toggle C OFF → toggle B OFF → toggle E OFF 3rd char = F → Check Load Meter. Acceptable range: 76%–82%. If load is in range: DONE. If load is outside range: Toggle breaker F.

3.4 RESOLUTION PROCEDURE

1	**Ask Fault Code**	Instruct Astronaut: 'Read the 3-character Fault Code on the EDS panel.'
2	**Ask current breaker states**	'Tell me which breakers currently have GREEN LEDs (ON) and which have RED FLASHING LEDs (TRIPPED).'
3	**Look up TABLE 3-A**	Using the FIRST character of the fault code, determine which breakers to toggle OFF.
4	**Look up TABLE 3-B**	Using the SECOND character, determine the toggle sequence.
5	**Give toggle instructions**	State clearly: 'Toggle [BREAKER] — that means flip the switch.' One breaker at a time.
6	**Check Load Meter**	'What percentage is the Load Meter showing?' Compare with TABLE 3-C row for 3rd character.
7	**Additional toggle if needed**	If load is out of range, instruct the additional toggle(s) from TABLE 3-C.
8	**Confirm resolution**	Module is solved when all LEDs stabilise GREEN and the Fault Code display shows '000'.

■ TRIPPED BREAKERS — IMPORTANT

If a breaker LED is FLASHING RED (tripped), it must be reset BEFORE it can be toggled. Instruct the Astronaut: 'Hold the [X] switch DOWN for 3 seconds, then release.' This resets the breaker to OFF state. Then proceed with normal toggle instructions.

■ LOAD METER NOTE The Load Meter reading changes AFTER each breaker toggle. Always read the FINAL load after completing all toggles from TABLE 3-A/3-B before consulting TABLE 3-C.

APP. APPENDIX — OPERATOR COMMUNICATION GUIDE

A.1 RECOMMENDED COMMUNICATION PROTOCOLS

Clear, standardised communication is the most critical factor in mission success. The following protocols are recommended for all Mission Control operators:

SITUATION	RECOMMENDED PHRASING
Requesting a reading	'Read me the [component] value.' — Avoid 'What does it say?' (ambiguous)
Giving a target	'Set [component] TO [value].' — Always say TO before the number
Confirming understanding	'Confirm: you will set [X] to [Y].' — Repeat back critical instructions
Astronaut reports confusion	'Stop. Take a breath. Describe only what you see directly in front of you.'
Time pressure	'We have [X] seconds. Focus on [component] only.' — One task at a time under pressure
Module solved	'Module secure. Move to [next module / next panel].'

A.2 PENALTY & ENERGY REFERENCE

EVENT	ENERGY EFFECT	ADDITIONAL CONSEQUENCE
Incorrect input on any module	−10%	Strike recorded + alarm sound
Module unsolved past time limit	−2% per second	Continuous drain until resolved
Module solved correctly	+5%	Efficiency bonus — confirm tone
3 strikes reached (Normal mode)	MISSION ABORT	Game over — mission failed
Energy reaches 0%	MISSION ABORT	Game over — all systems lost

■ MISSION SUCCESS CRITERIA

All critical modules resolved before timer expires AND Energy > 0% AND Strikes < maximum. Upon successful completion, the Astronaut will receive atmospheric re-entry clearance and the mission is recorded as complete. Godspeed, Mission Control.